Benedict Aaron Tjandra

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Google Scholar

Education

University of Oxford, St. John's College — Oxford, United Kingdom

Master of Science in Advanced Computer Science — 2023 - 2024

Supervisors: **Prof. Michael Bronstein** (Independent, uncredited project), **Prof. Yarin Gal** (Thesis).

Independent project accepted to *NeurIPS Symmetry and Geometry in Neural Representations 2024 (Oral)*. Thesis accepted to *NeurIPS Safe Generative AI Workshop 2024*.

University of Cambridge, Trinity Hall — Cambridge, United Kingdom

Bachelor of Arts (Hons) in Computer Science, Natural Sciences — 2017 - 2020

Grade: Double First - 86.75%, Rank: 1 / 102 (**Valedictorian**)

Awards: Winifred Georgina Holgate-Pollard Memorial Prize, Lee-Yung Prize for Computer Science, Bateman Scholarship, Trinity Hall Scholarship Award (2018, 2019)

Work Experience

Google DeepMind — London, United Kingdom

Senior Research Engineer — Sept 2025 - Present

- Lead for a Gemini pre-training modelling research effort. Currently showing up to +30% performance gains.
- Lead of SkillSmith – a SOTA method to compose text and learned skills (KV-caches) for agents. Achieved up to +17% (trained) and +160% gains over strongest baselines. Spearheaded publication, resulting in a patent and a paper.
- Led theoretical research proving a fundamental trade-off between information propagation speed and quality in computational graphs. Spearheaded publication leading to a paper.
- Built key infrastructure for a long-running agent to continually learn, and led experimentation to show early self-improvement capabilities.

XTX Markets — London, United Kingdom

Core Developer — Sept 2021 - Sept 2025

- Research on portfolio optimisation methods and FX performance attribution.
- Developed an intraday analysis framework from scratch to power historical and real-time risk calculations, which include high-frequency performance attributions, stress tests, and risk monitoring systems.
- Main contributor to an in-house distributed Kafka-based micro-services architecture that handles the firm's middle-back office, e.g. real-time trade reconciliation, enrichment, capture, surveillance, risk and position calculation systems, DevOps, and real-time pricing systems.

Amazon Web Services — Cambridge, United Kingdom

Software Development Engineer I — Oct 2020 - Aug 2021

- Delivered the launch of S3 Object Lambda – a serverless solution to bring compute closer to storage.
- Maintained and improved the scalability and reliability of highly-distributed, global cloud services serving >100 TPS – S3 Select & S3 Object Lambda.

Goldman Sachs — London, United Kingdom

Summer Technology Analyst — Jul 2020 - Aug 2020

- SecDB Architecture: Optimised trade serialisations using C++ improving runtime performance by 50%.

Utterberry LTD — London, United Kingdom

Software Engineering Intern (AI Development) — Jun 2019 - Sep 2019

- Developed neural-based applications to perform facial recognition and estimate traffic density. Additionally developed an in-house backend infrastructure to deploy and execute jobs and scripts on internal devices.

Research Experience

University of Cambridge — Cambridge, United Kingdom (Remote)

Visiting Researcher, *Machine Learning Group* — April 2025 - September 2025

- Supervisor: **Prof. José Miguel Hernández-Lobato**
- Molecular diffusion on latent Fourier space.

EMBL-EBI Cambridge, United Kingdom
Visiting Researcher, *Saez-Rodriguez Lab* July 2025 - September 2025
- Supervisor: **Dr. Aurelien Dugord**
- Large Language Models for denoising and inference on gene regulatory networks.

Montreal Institute of Learning Algorithms (MILA) Montreal, Canada (Remote)
Visiting Researcher, *Department of Computer Science* April 2025 - September 2025
- Supervisor: **Prof. Guillaume Rabusseau**
- Designed a synthetic benchmark to evaluate the reasoning capabilities of temporal graph neural networks.

University of Oxford Oxford, United Kingdom
Research Assistant, *Department of Computer Science* February 2024 - December 2024
- Supervisor: **Prof. Michael Bronstein**
- Proposed to add source-target identification to Temporal Graph Networks (TGNs), and subsequently showed that it significantly boosts its expressivity, making it outperform all current temporal graph methods for dynamic node affinity prediction.
- Independently came up with the idea, performed all experiments, did all of the proofs, wrote most of the paper, and presented the work as an oral presentation at NeurIPS.

University of Oxford Oxford, United Kingdom
Research Assistant, *Oxford Applied & Theoretical ML (OATML) Group* February 2024 - October 2024
- Supervisor: **Prof. Yarin Gal**
- Proposed to leverage semantic entropy to fine-tune LLMs to abstain from questions beyond their understanding. Our method achieves competitive results on short and long-form answering settings without requiring ground-truth labels.
- Main originator of the idea, performed all experiments, and wrote most of the workshop paper published at NeurIPS.

University of Cambridge Cambridge, United Kingdom
Undergraduate Research Assistant, *Department of Computer Science* December 2019 - June 2020
- Supervisor: **Prof. Pietro Liò**
- Undergraduate research project exploring the feasibility of integrating graph neural networks to improve the reasoning of Visual Question Answering (VQA) models.

Publications

SkillSmith: Learning to Compose Parametric Skills and Textual Knowledge
Lucio M. Dery*, Benedict Aaron Tjandra*, Siavash Samiei, Adhiguna Kuncoro, Zohar Yahav, Arthur Szlam, and Jiajun Shen. Link.

Appeared at COLM LLA Workshop 2026.

Feedforward Mixing is as Sharp as it is Slow in Reverse
Benedict Aaron Tjandra, Avi Wigderson, João GM Araújo, Alex Vitvitskyi, Federico Barbero, and Petar Veličković. Link.

Appeared at ICML FoGen and GFM Workshops 2026.

T-GRAB: A Synthetic Diagnostic Benchmark for Learning on Temporal Graphs
Alireza Dizaji, Benedict Aaron Tjandra, Mehrab Hamidi, Shenyang Huang, and Guillaume Rabusseau. Link.

Appeared at KDD 2026 (Oral).

Enhancing the Expressiveness of Temporal Graph Networks via Source-Target Identification
Benedict Aaron Tjandra, Federico Barbero, and Michael Bronstein. Link.

Appeared at NeurIPS Symmetry and Geometry of Neural Representations Workshop 2024 (Oral).

Fine-Tuning Large Language Models to Appropriately Abstain via Semantic Entropy
Benedict Aaron Tjandra, Muhammed Razzak, Jannik Kossen, and Yarin Gal. Link.

Appeared at NeurIPS Safe Generative AI Workshop 2024.

On Graph Neural Network Ensembles for Large-Scale Molecular Property Prediction
Edward Elson Kosasih, Joaquin Cabezas, Xavier Sumba, Piotr Bielak, Kamil Tagowski, Kelvin Idanwekhai, Benedict Aaron Tjandra, and Arian Rokkum Jamasb. Link.

Standalone arXiv paper.